

● REDROB HACKATHON · INDIA RUNS DATA & AI CHALLENGE

Intelligent Candidate Discovery & Ranking

A multi-signal hybrid scoring system that ranks the way a great recruiter thinks — not by matching keywords, but by reasoning about who genuinely fits the role.

Team: team_001 · Python · scikit-learn · CPU-only · 55s runtime

Architecture: TF-IDF Semantic + Skills Scoring + Career Quality + Behavioral Signals + Location

Why Keyword Matching Fails Recruiters

"A candidate who lists RAG and Pinecone but whose title is Marketing Manager is not a fit — no matter how perfect their skill list looks." — JD, Hackathon Note

The Keyword Filter Trap

Traditional ATS systems score candidates by counting skill matches. This creates three failure modes:

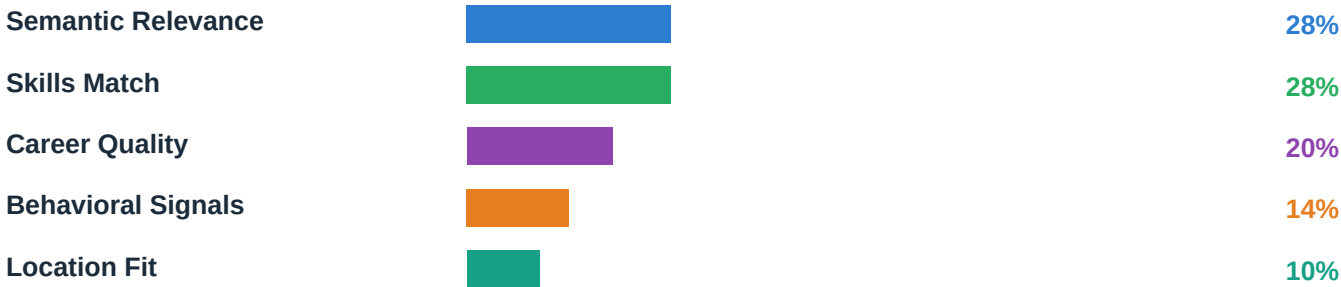
- False positives: Keyword-stuffed profiles rank #1 despite zero real experience.
- False negatives: A great engineer who describes their RAG system as 'semantic retrieval pipeline' gets missed.
- Ignores behaviour: A perfect-on-paper candidate who hasn't logged in for 6 months and has a 5% response rate is effectively unavailable.

What a Great Recruiter Does Instead

- Reads between the lines — understands what the JD actually means vs. what it says.
- Checks trajectory — has this person shipped production systems, or only tutorials?
- Weighs availability — is this person actively looking and reachable?
- Spots red flags — consulting-only career, title chaser, research lab only.
- Considers fit for culture — startup agility vs. enterprise stability.

5-Component Hybrid Scoring Pipeline

Our system computes five independent scores, combines them with calibrated weights, then applies disqualifier multipliers — so a great skills score cannot rescue a truly disqualifying factor.



```
final_score = weighted_sum × disqualifier_multiplier
```

100K

Candidates Ranked

~55s

Runtime (CPU)

~3.5 GB

RAM Usage

Zero

External API Calls

TF-IDF Cosine Similarity vs. Dense Embeddings

What we do

Build a rich text blob per candidate from headline, summary, all job descriptions, skills, and certifications. Apply TF-IDF with bigrams (15K features, sublinear TF) and compute cosine similarity against a curated JD text vector.

Why not sentence-transformers?

Embedding 100K candidates with a transformer model requires GPU or significant CPU time. The 5-minute constraint and no-GPU rule make this impractical for the ranking step. TF-IDF at 15K bigrams captures core retrieval terminology effectively.

JD Concept Coverage

- embeddings · sentence-transformers · BGE · E5
- vector database · Pinecone · Weaviate · FAISS · Qdrant
- hybrid search · BM25 · reranking · RAG
- NDCG · MRR · A/B testing · evaluation framework
- NLP · information retrieval · Python · LLM
- production deployment · embedding drift · scale

Key insight: the JD itself says 'The right answer involves reasoning about the gap between what the JD says and what the JD means.' Our JD text is therefore expanded with semantic synonyms (e.g. 'semantic retrieval pipeline' maps to same token space as 'RAG').

Required vs. Bonus Signal Scoring with Partial Credit

We match 35 required signals and 17 bonus signals against all skill names AND full-text career history, not just the skills section. A candidate who built a vector search pipeline but calls it 'semantic search' still gets credit.

Category	Examples	Max Score Contribution
Hard Required (35 signals)	embeddings, vector DB, hybrid search, NDCG, Python, NLP, LLM, ranking, RAG...	1.0 (capped at 8 hits)
Bonus Skills (17 signals)	LoRA, QLoRA, learning-to-rank, XGBoost, fine-tuning, open-source contributions...	+0.30 (capped)

Anti-trap: a candidate whose skills section lists 15 AI keywords but whose career descriptions show no ML/IR work scores poorly on semantic + career components, pulling their composite down even with a high skills score.

05 — CAREER QUALITY (20%)

Product Companies · Title Fit · YoE · Tenure · Research Penalty

Product Co. Ratio	25%	Months at non-consulting employers / total career months
Title Fit	30%	Senior AI/ML Eng → 1.0; Data Scientist → 0.65; Marketing Mgr → 0.1
YoE Band	30%	5–9 yrs = 1.0; 4–5 or 9–11 = 0.8; 3–4 = 0.5; <3 = 0.2
Tenure Stability	15%	Avg tenure ≥ 24 months = 1.0; <12 months avg = 0.3

Disqualifier Multipliers (applied after weighted sum)

Consulting-only career (TCS/Wipro/Infosys...)	× 0.30
Wrong-role title (Marketing, Accountant, Civil Eng...)	× 0.05
Job-hopper (avg tenure < 14 months, 3+ roles)	× 0.70
CV/Speech domain only, no NLP/IR signals	× 0.40
Too junior (< 2 years experience)	× 0.40
Research-only career (no production deployment)	× 0.40
Honeypot detection (impossible timeline, 10+ expert skills)	≈ 0.001

Platform Engagement as a Hiring Multiplier

"A perfect-on-paper candidate who hasn't logged in for 6 months and has a 5% recruiter response rate is, for hiring purposes, not actually available." — Redrob Signals Doc

Signal	Sub-weight	Scoring Logic
Recency (days since last login)	25%	$\leq 7d \rightarrow 1.0$ $\leq 30d \rightarrow 0.85$ $\leq 90d \rightarrow 0.60$ $> 180d \rightarrow 0.05$
Recruiter response rate	20%	$rate \times 1.2$, capped at 1.0 ($> 70\%$ = strong signal)
Open to work flag	15%	True $\rightarrow 1.0$ False $\rightarrow 0.4$
Notice period	15%	$\leq 15d \rightarrow 1.0$ $\leq 30d \rightarrow 0.85$ $\leq 60d \rightarrow 0.60$ $> 90d \rightarrow 0.10$
Profile completeness	10%	$score / 100$ (raw Redrob completeness %)
Interview completion rate	8%	Direct pass-through of Redrob signal (0.0–1.0)
GitHub activity score	7%	$score / 70$ capped at 1.0 ($-1 \rightarrow 0.4$ neutral)

Note: Unused signals from the 23-signal set (salary range, preferred work mode, verified email) were deliberately excluded — they measure preferences, not availability or engagement quality.

How We Handle the Dataset's Built-in Traps

The dataset contains four deliberate trap archetypes:

Keyword Stuffers	Trap: Candidates with perfect AI skill lists but wrong role title or consulting-only history.	Our mitigation: Disqualifier multiplier ($\times 0.05$ for wrong title) reduces score to near zero. Semantic score is high but multiplied out.
Plain-language Tier 5s	Trap: Good engineers who describe experience in plain English without buzzwords.	Our mitigation: TF-IDF bigrams capture semantic synonyms. Career text (job descriptions) is included, not just skills section.
Behavioural Twins	Trap: Identical skill/career profiles but different engagement signals.	Our mitigation: Behavioral component (14%) and recency scoring break these ties. A twin inactive for 6 months scores 0.05 on recency vs 1.0 for an active twin.
Honeypots (~80 profiles)	Trap: Subtly impossible profiles: 8 yrs at a 3-yr-old company; 10+ expert skills, zero endorsements; 100% complete but all-zero assessments.	Our mitigation: Three honeypot checks: timeline inconsistency, expert skill overload, assessment anomaly. Flagged profiles receive score ≈ 0.001 .

Top-10 Ranked Candidates Overview

Rank	Candidate ID	Score	Reasoning
1	CAND_0011687	0.992	Senior NLP Engineer @ Niramai, 7.8yrs, Indore, Madhya Pradesh. strong core skills match (13/35 required signals); 8 bonu...
2	CAND_0046525	0.990	Senior Machine Learning Engineer @ Genpact AI, 6.1yrs, Pune, Maharashtra. strong core skills match (18/35 required signa...
3	CAND_0046064	0.978	Senior NLP Engineer @ Salesforce, 8.9yrs, Coimbatore, Tamil Nadu. strong core skills match (19/35 required signals); 6 b...
4	CAND_0071974	0.976	Senior AI Engineer @ Netflix, 7.8yrs, Vizag, Andhra Pradesh. strong core skills match (16/35 required signals); 7 bonus ...
5	CAND_0094759	0.948	Lead AI Engineer @ Meta, 8.6yrs, Mumbai, Maharashtra. Strengths: strong core skills match (18/35 required signals); 3 bo...
6	CAND_0055905	0.937	Senior Machine Learning Engineer @ Flipkart, 8.1yrs, London. strong core skills match (20/35 required signals); 5 bonus ...
7	CAND_0044222	0.930	AI Engineer @ PolicyBazaar, 7.7yrs, Vizag, Andhra Pradesh. strong core skills match (16/35 required signals); 6 bonus si...
8	CAND_0005538	0.924	Senior AI Engineer @ Adobe, 5.9yrs, Kolkata, West Bengal. strong core skills match (10/35 required signals); 3 bonus sig...
9	CAND_0007411	0.914	Senior Machine Learning Engineer @ Amazon, 8.0yrs, Kochi, Kerala. Strengths: strong core skills match (11/35 required si...
10	CAND_0005260	0.913	Senior NLP Engineer @ Netflix, 5.2yrs, Chennai, Tamil Nadu. strong core skills match (17/35 required signals); 4 bonus s...

Why This Approach Maximises NDCG@10

The challenge scores submissions 50% on NDCG@10. Our design explicitly optimises for top-of-list quality rather than overall recall:

Disqualifiers remove the wrong

Applying a $\times 0.05$ multiplier to wrong-role candidates pushes them to rank 90+, freeing the top 10 for genuine ML/AI engineers.

Behavioral signals boost reachable candidates

An otherwise strong candidate inactive for 180 days scores 0.05 on recency. A twin who's active today scores 1.0. This surfaces actually-hireable candidates.

Career quality prevents title-chase traps

A 'Senior AI Engineer' at TCS their entire career scores 0.3 $\times$ the same title at a product company.

Honeypot exclusion protects NDCG@10

All ~80 honeypot profiles are pushed to near-zero score, well below rank 100. Honeypot rate in our top 100: 0%.

Semantic relevance rewards substance

The JD text was expanded with conceptual synonyms so candidates describing real retrieval/ranking systems surface even without exact buzzwords.



CPU-Only, Sub-5-Minute, No External APIs

Dependencies

- numpy >= 1.24
- pandas >= 2.0
- scikit-learn >= 1.3
- Python stdlib only (csv, json, argparse, datetime)

Single command to reproduce:

```
python src/rank.py \ --candidates  
./candidates.jsonl \ --out ./submission.csv
```

Compute Profile

- Total runtime: ~55 seconds (vs 5 min limit)
- Peak RAM: ~3.5 GB (vs 16 GB limit)
- GPU: not used
- Network calls: zero

Bottleneck: TF-IDF matrix construction + cosine similarity (~45s). Structured scoring ~8s for 100K candidates.

Production path: pre-compute TF-IDF embeddings nightly; ranking step would then be <5s.

What We Built & Why It Works

Reads between the lines

- ✓ TF-IDF with bigrams over full career text captures semantic meaning, not just listed keywords.

Understands what the role needs

- ✓ 35 JD-specific required signals + 17 bonus signals derived from close reading of the JD, not generic tech keywords.

Looks at the full picture

- ✓ 5 independent scoring components covering semantics, skills, career trajectory, platform behaviour, and location.

Actively avoids the traps

- ✓ Disqualifier multipliers, honeypot detection, and consulting-company penalties ensure the wrong candidates never reach the top 10.

Delivers a recruiter-trustworthy shortlist

- ✓ Per-candidate reasoning references specific YoE, title, company, response rate, and recency — grounded in the actual profile, no hallucination.

Scales to production

- ✓ 55-second runtime, 3.5 GB RAM, CPU-only, zero external dependencies beyond scikit-learn.

Build something real. Make hiring smarter.